

Lecture 11: The Derivative of $\ln x$

GOAL: Differentiate $\ln x$ and introduce the technique of Logarithmic Differentiation. Finish the proof of the Power Rule.

Summary:

$$\begin{aligned} [\ln x]' &= \frac{1}{x} & [\log_b x]' &= \frac{1}{x \ln b} \\ [\ln f(x)]' &= \frac{f'}{f} & x^p &= e^{p \ln x} \end{aligned}$$

